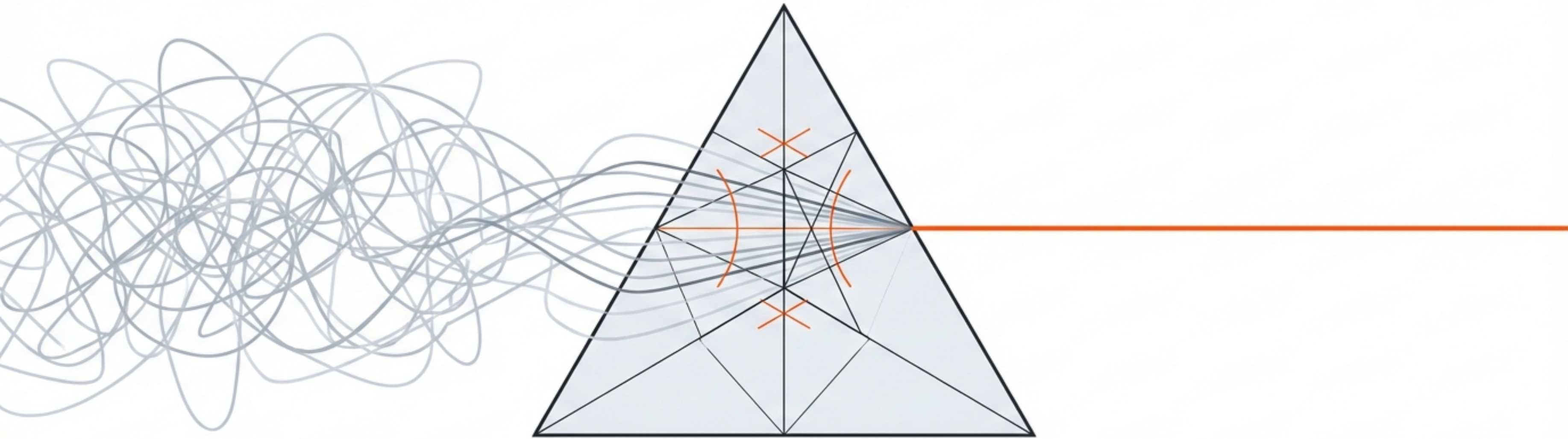
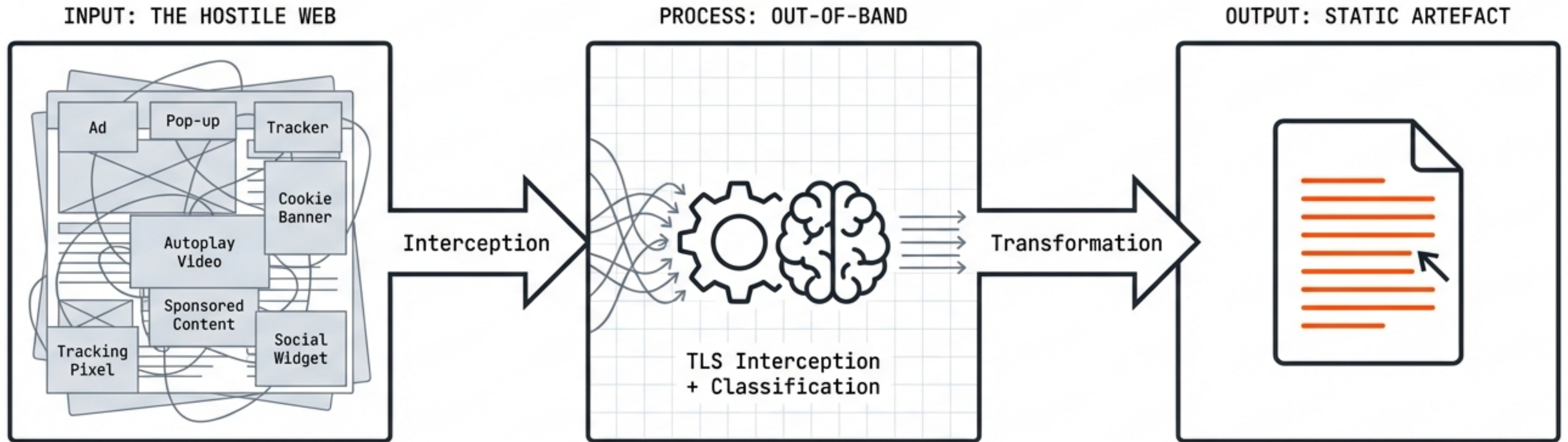


Static Page Transformation MVP

A Research-Based Approach to Reducing Information Overload via mitmproxy



Decoupling Content Discovery from Content Consumption



Core Concept

A developer interception proxy that serves static, pre-transformed HTML. The heavy lifting happens out-of-band.

The Mechanism

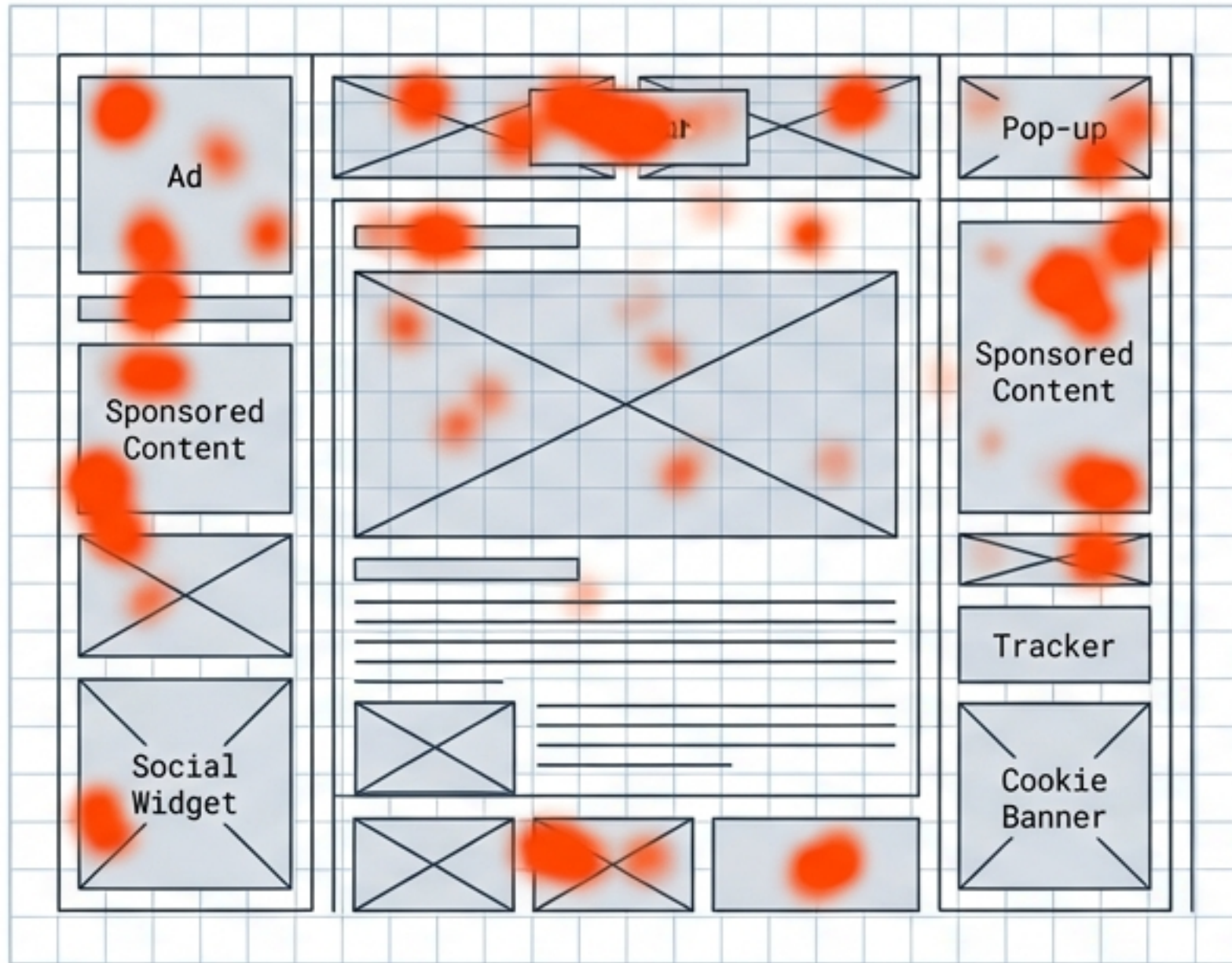
Moves classification and rewriting to the backend, ensuring the runtime browsing experience remains predictable.

Value Proposition

Aligns browsing with information overload research by serving stable, signal-rich assets (S3 Pattern).

The High Cost of the Hostile Web

The Reality: Scattered Attention



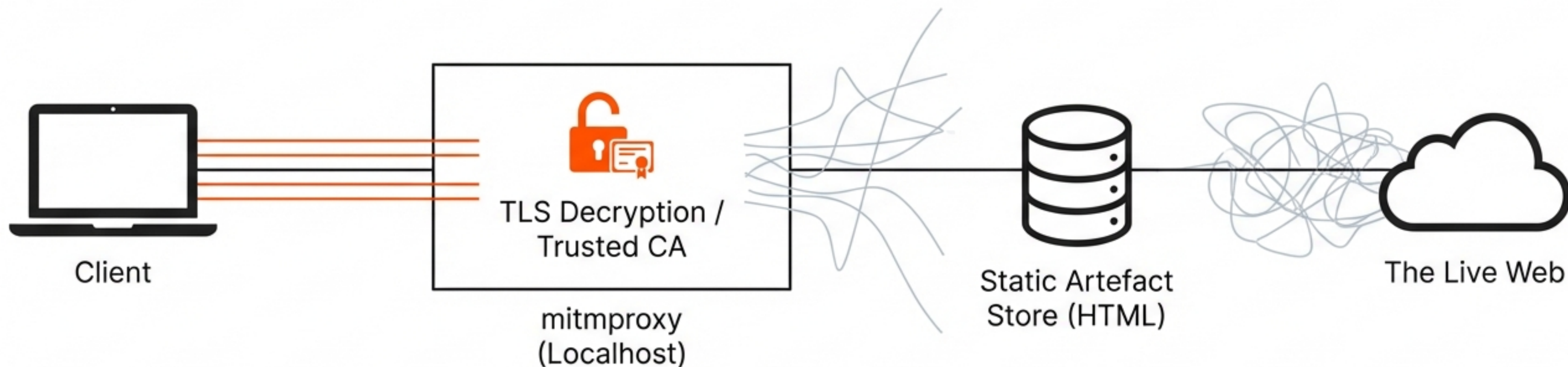
Source: Jakob Nielsen, "How Users Read on the Web"

Research Context

- **Information Overload**
Impacts decision-making, productivity, and wellbeing.
Intl Journal of Information Management, 2024
- **Scanning Behaviour**
Users do not read linearly; they scan for keywords. Promotional layouts break this flow.
Nielsen, Norman Group
- **Cognitive Barriers**
 - Banner Blindness: Users ignore elements that resemble ads.
 - Intrusive Interruptions: Pop-ups drive abandonment.

Coalition for Better Ads

Architecture: The Static Serving Model



TLS Interception

Decrypts traffic on the fly using a locally trusted CA certificate. This allows deep inspection and replacement of content.

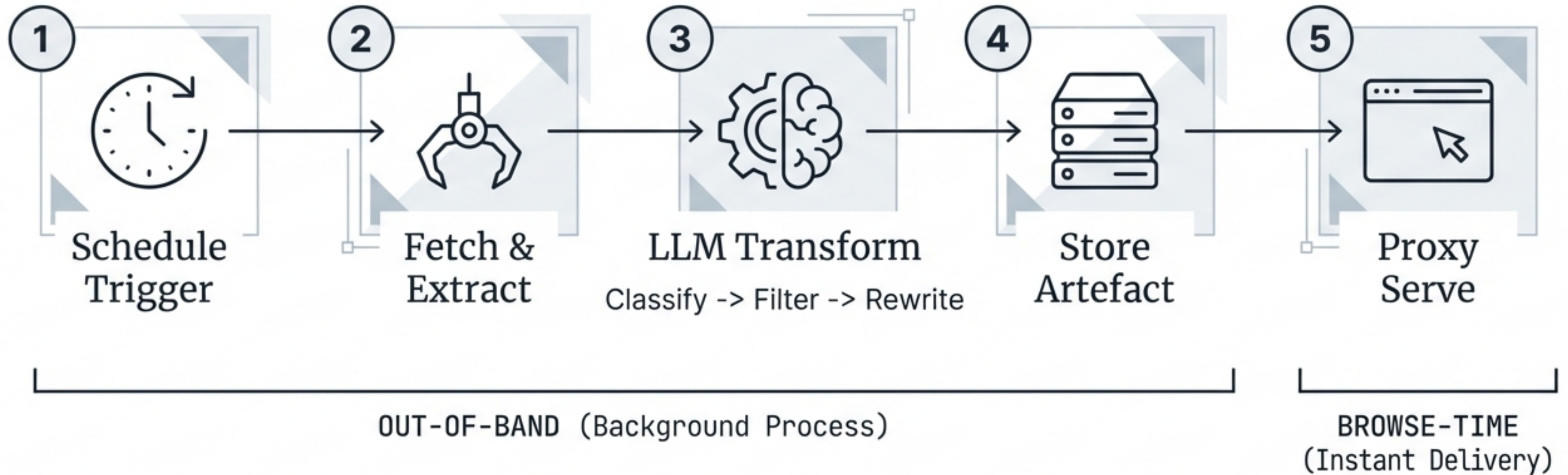
The Trust Model

Front-loaded Friction: The user installs the cert once during onboarding, enabling friction-free daily use.

Target Scope

IN-SCOPE: Stable content (Docs, Blogs, Announcements).
OUT-OF-SCOPE: Dynamic feeds (YouTube, LinkedIn).

The Transformation Pipeline



Key Differentiator: The user never waits for the LLM. They only see the pre-computed result.

Scenario A: The Focus Lens Single Page Transformation

User Story

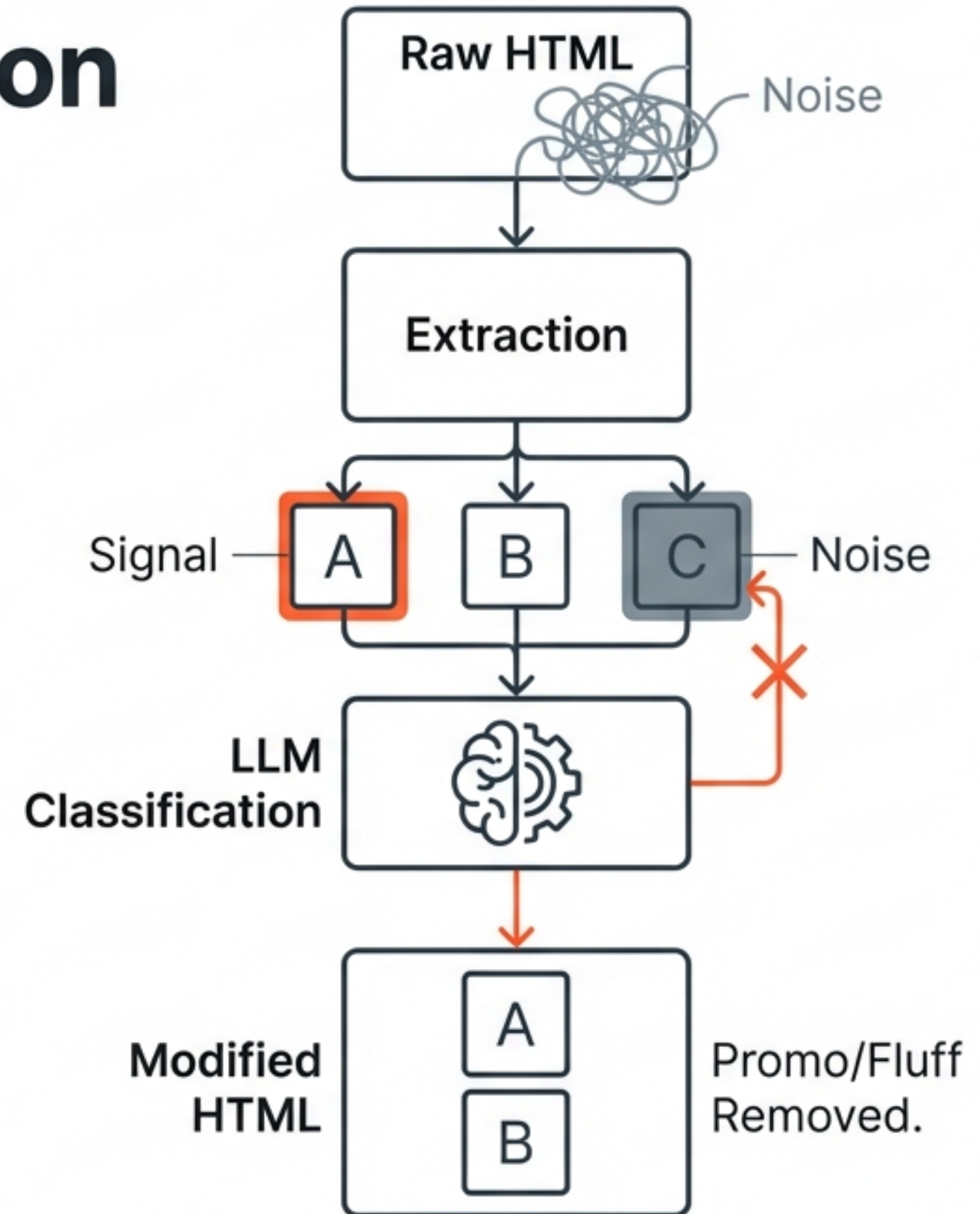
I want to see only factual blocks; hide the promotional noise.

Research Backing

Concise, objective writing improves usability (Morkes & Nielsen).

Success Metric

Faster identification of “worth opening” items.



Scenario B: The Consistent Section Lens



User Story

I want the same filtering rules applied across a whole documentation set to avoid cognitive context switching.

The Value: Predictability

If the “Lens” changes per page, the user must re-learn the layout. Consistent filtering supports learned scanning patterns.

Risk Mitigation

Transparent toggles to view original content prevent the “Filter Bubble” effect.

Scenario C: Role-Based Filtering

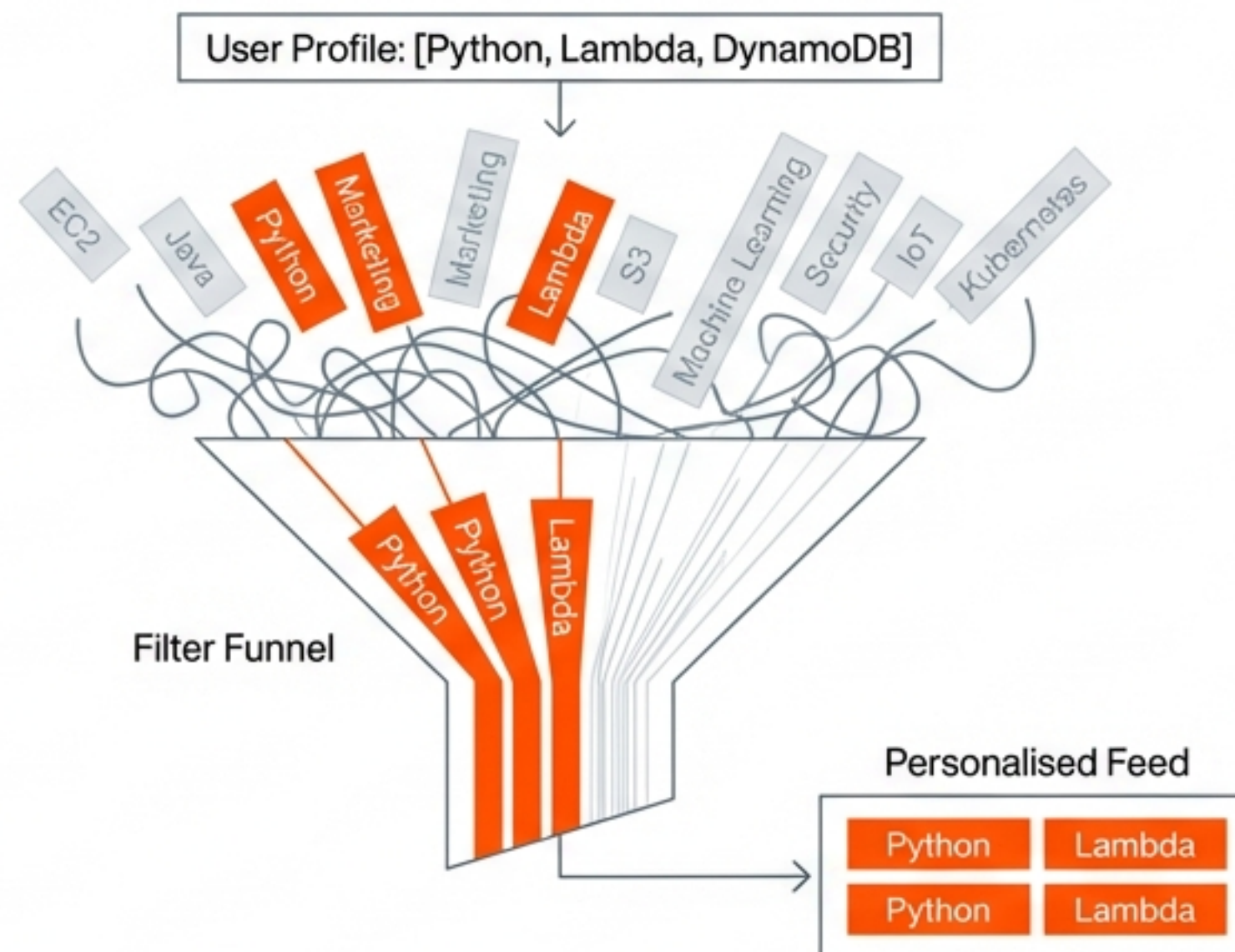
Case Study: AWS re:Invent Announcements

User Story

Engineering Manager: “I need to see only updates relevant to my stack (e.g., Python + Serverless).”

Why it Fits

High volume, broad content. The canonical “overload” environment.



Success: Answer “What changed for us?” in seconds.

Scenario D: Nuisance Neutralisation

User Story

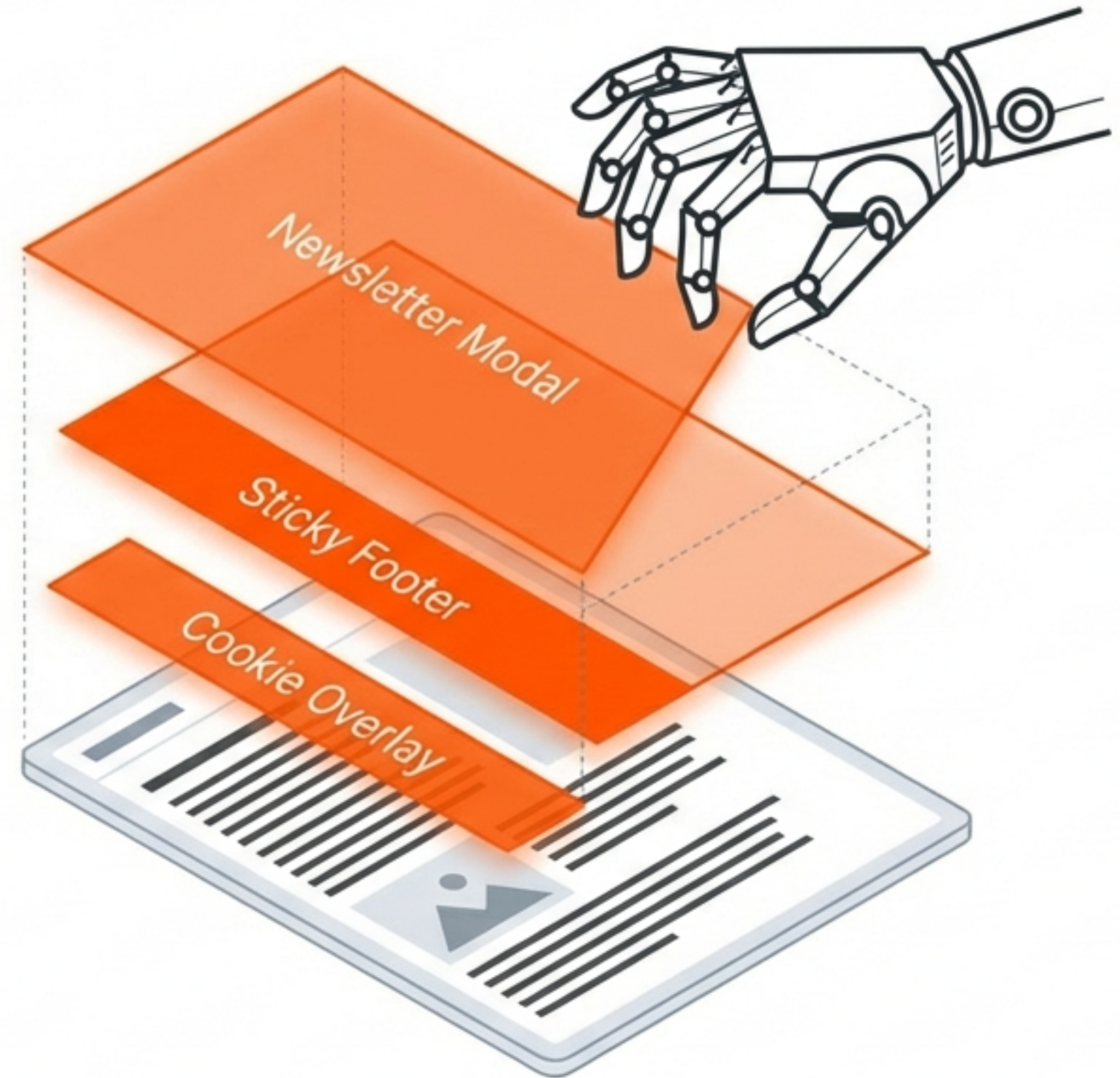
Read without pop-ups, overlays, or sticky banners.

Research Backing

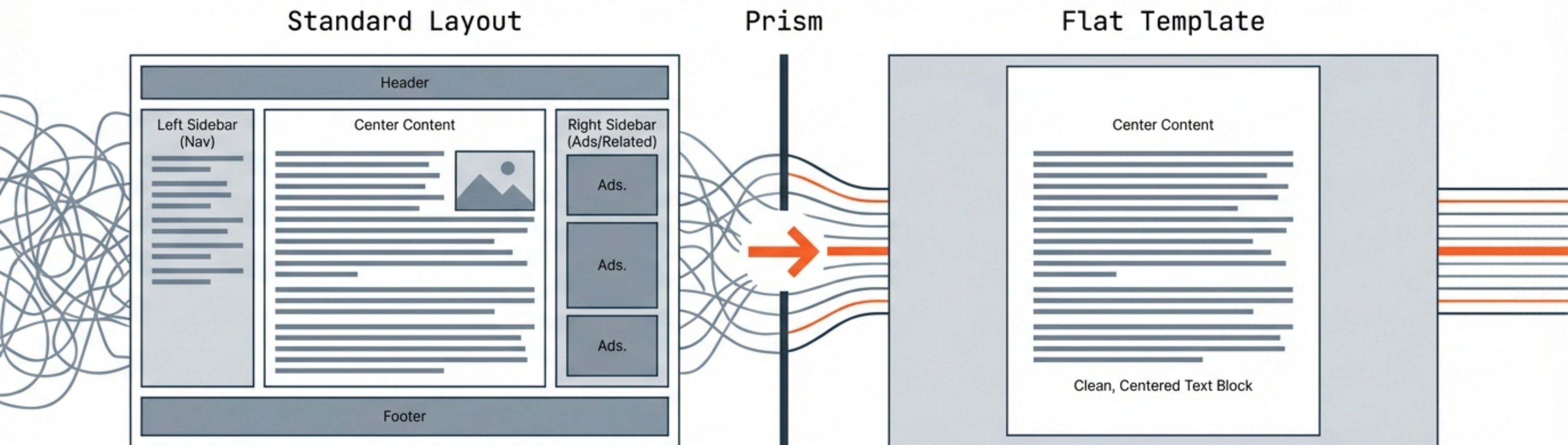
Coalition for Better Ads: Interruption-based elements drive abandonment and are “below-threshold” experiences.

The Workflow

Identify patterns (modals, z-index overlays) -> Neutralise in HTML -> Store “Clean” version.



Scenario E: The Flat UI Transformation



User Story: Simplified layout for easier reading and copy-pasting.

Research Backing: Nielsen's F-shaped reading pattern; GOV.UK content principles favor content flattening.

Differentiation: This is not just ad-blocking; it is a structural redesign of the page.

Feasibility: Governance & Trust Models



The Challenge: Break-and-Inspect

Interception requires the client to trust the proxy's CA. This is a high-friction trust model.

The Challenge: Break-and-Inspect

Interception requires the client to **trust** the proxy's **CA**. This is a high-friction trust model.

The Enterprise Context

Organisations already use **Secure Web Gateways** (Gartner) for policy enforcement. This MVP is '**functional adjacency**'—using the same tech for Experience rather than Security.

Guidance

NIST SP 1800-37: Visibility into TLS 1.3.

Key Question

Will users accept this trust model on personal devices, or is this strictly an enterprise tool?

Validating Market Pull: The Research Agenda

Focus Lens

Do users want removal (risk of missing out) or just de-emphasis?



Role Filtering

Centralized (Manager-defined) profiles or User-defined?



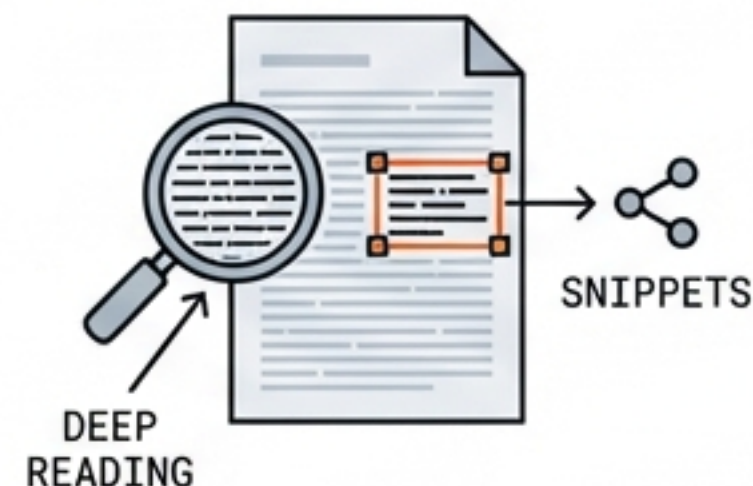
Nuisance

Which format triggers the most rage? (Pop-ups vs. Autoplay vs. Sticky Banners).



Flat UI

Is the goal deep reading, or extracting snippets for sharing?



Reclaiming the Signal



Summary

By moving the heavy transformation logic **out-of-band**, we preserve the speed of the web while fixing its signal-to-noise ratio.

Strategic Outlook

Technology (mitmproxy + LLM) is ready. **Research** (Nielsen, Norman) confirms the need.

Next Step

Prototype the '**AWS re:Invent Announcement**' scenario. A canonical example of high-volume, low-signal content.